

# TECHNICAL REPORT

## Customer Churn Analysis

### 1. Introduction

Customer churn is a major concern for subscription-based businesses, since retaining an existing customer is generally cheaper than acquiring a new one. For this project, I analyzed a dataset of 500 customers to understand what factors are associated with churn and to provide the business with some practical, data-backed recommendations.

### 2. Data Description

The dataset (customer\_churn.csv) contains 500 customer records with 9 columns:

- CustomerID — unique identifier for each customer
- Tenure — number of months the customer has been with the company
- MonthlyCharges — amount billed monthly
- TotalCharges — total amount billed to date
- Contract — type of contract (Month-to-month, One year, Two year)
- PaymentMethod — how the customer pays
- PaperlessBilling — whether billing is paperless (Yes/No)
- SeniorCitizen — whether the customer is a senior citizen (0/1)
- Churn — whether the customer left (1) or stayed (0)

On checking the data, I found no missing values or duplicate rows, so minimal cleaning was needed before analysis.

### 3. Methodology

I approached this project in three stages:

#### 3.1 Data Cleaning

Verified data quality — checked for missing values, duplicates,

and confirmed data types were correct.

## 3.2 Exploratory Data Analysis (EDA)

Created visualizations to understand churn patterns across different customer segments, including contract type, tenure, and monthly charges.

## 3.3 Statistical Analysis

Used correlation analysis and hypothesis testing (t-test) to verify whether the patterns observed in the EDA stage were statistically significant, rather than just random variation.

# 4. Analysis & Results

## 4.1 Overall Churn Rate

Of the 500 customers, 10.6% churned during the period covered by this dataset.

## 4.2 Churn by Contract Type

Churn rates varied significantly by contract type:

- Month-to-month: 20.6%
- One year: 4.3%
- Two year: 6.9%

Customers on flexible, short-term contracts are far more likely to churn than those on longer commitments.

## 4.3 Tenure Analysis

Churned customers had an average tenure of 6 months, compared to 40 months for retained customers. This suggests that churn risk is highest early in the customer relationship.

## 4.4 Correlation Analysis

A correlation heatmap was used to examine relationships between numerical variables. Tenure showed the strongest correlation with churn (-0.51), indicating that longer-tenured customers are considerably less likely to churn. Monthly charges showed a weaker positive correlation (0.11) with churn.

## 5. Statistical Testing

To confirm that the tenure difference between churned and retained customers was not due to random chance, I ran an independent t-test:

- T-statistic: -13.20
- P-value: < 0.001

Since the p-value is well below the standard threshold of 0.05, this difference is statistically significant, confirming that tenure is a genuine factor associated with churn rather than coincidence.

## 6. Conclusion & Recommendations

The analysis shows that churn is not random — it is strongly associated with contract type and customer tenure. Based on these findings, I would recommend:

1. Prioritizing retention efforts for customers in their first 6-12 months, as this is the highest-risk period
2. Offering incentives to shift customers from month-to-month to annual contracts
3. Monitoring customers with higher monthly charges for potential engagement or loyalty offers
4. Designing targeted retention campaigns specifically for new, month-to-month customers

These steps, if implemented, could help the business reduce churn in its highest-risk customer segments.